

Derivation of Free Cash Flow, Weighted Average Cost of Capital, and Adjusted Present Value

Chapter 10 demonstrated numerically the equivalence of enterprise discounted cash flow (DCF), adjusted present value (APV), and the cash-flow-to-equity valuation when leverage (as measured by the market-based debt-to-equity ratio) is constant. This appendix derives the key terms in each model—namely, free cash flow (FCF) and the weighted average cost of capital (WACC)—and demonstrates their equivalence algebraically.

To simplify the analysis, we assume cash flows to equity are growing at a constant rate, g . This way we can use growth perpetuities to analyze the relationship between methods.¹

ENTERPRISE DISCOUNTED CASH FLOW

By definition, enterprise value (V) equals the market value of debt (D) plus the market value of equity (E):

$$V = D + E$$

¹ For an analysis that applies to more complex situations (i.e., when cash flows can follow any pattern), see J. A. Miles and J. R. Ezzell, "The Weighted Average Cost of Capital, Perfect Capital Markets, and Project Life: A Clarification," *Journal of Financial and Quantitative Analysis* 15 (1980): 719–730 (for a discussion of enterprise DCF and WACC); and S. C. Myers, "Interactions of Corporate Financing and Investment Decisions: Implications for Capital Budgeting," *Journal of Finance* 29 (1974): 1–25 (for a discussion of adjusted present value).

To examine the components of enterprise value, multiply the right side of the equation by a complex fraction equivalent to 1 (the numerator equals the denominator, an algebraic trick we will use many times):

$$V = (D + E) \left(\frac{D(1 - T_m)k_d + CF_e - D(g)}{D(1 - T_m)k_d + CF_e - D(g)} \right) \quad (\text{B.1})$$

where T_m = marginal tax rate

k_d = cost of debt

CF_e = cash flow to equity holders

g = growth in cash flow to equity holders

Over the next few steps, the fraction's numerator will be converted to free cash flow (FCF). We will show later that the denominator equals the weighted average cost of capital. Start by defining the numerator as FCF:

$$\text{FCF} = D(1 - T_m)k_d + CF_e - D(g)$$

If the market value of debt equals the face value of debt, the cost of debt will equal the coupon rate, and D times k_d will equal the company's interest expense. Therefore,

$$\text{FCF} = \text{Interest}(1 - T_m) + CF_e - D(g)$$

By definition, cash flow to equity (CF_e) equals earnings before interest and taxes (EBIT) minus interest, taxes, and net investment, plus the increase in debt. Assuming the ratio of debt to equity is constant, the annual increase in debt will equal $D(g)$. Why? Since cash flows to equity are growing at g , the value of equity also grows at g . Since the ratio of debt to equity remains constant (a key assumption), the value of debt must also grow at g . Substitute the definition of cash flow to equity into the preceding equation:

$$\text{FCF} = \text{Interest}(1 - T_m) + \text{EBIT} - \text{Interest} - \text{Taxes} - \text{Net Investment} + D(g) - D(g)$$

Next, distribute the after-tax interest expression into its two components, and cancel $D(g)$:

$$\text{FCF} = \text{Interest} - T_m(\text{Interest}) + \text{EBIT} - \text{Interest} - \text{Taxes} - \text{Net Investment}$$

Simplify by canceling the interest terms and rearranging the remaining terms:

$$\text{FCF} = \text{EBIT} - [\text{Taxes} + T_m(\text{Interest})] - \text{Net Investment}$$

Chapter 11 defines operating taxes as the taxes a company would pay if the company were financed entirely with equity. Operating taxes therefore equal reported taxes plus the interest tax shield (as interest is eliminated, taxes would rise by the interest tax shield). This leads to the definition of free cash flow we use throughout the book:

$$\text{FCF} = \text{EBIT} - \text{Operating Taxes} - \text{Net Investment}$$

Next, we focus on the denominator. To derive the weighted average cost of capital (WACC), start with Equation B.1, and multiply CF_e by 1, denoted as $(k_e - g)/(k_e - g)$:

$$V = (D + E) \left(\frac{\text{FCF}}{D(1 - T_m)k_d + \frac{\text{CF}_e}{k_e - g}(k_e - g) - D(g)} \right)$$

where k_e = cost of equity.

If equity cash flows are growing at a constant rate, the value of equity equals CF_e divided by $(k_e - g)$. Therefore, the growing perpetuity in the denominator can be replaced by the value of equity (E) and distributed:

$$V = (D + E) \left(\frac{\text{FCF}}{D(1 - T_m)k_d + E(k_e) - E(g) - D(g)} \right)$$

In the denominator, collapse $E(g)$ and $D(g)$ into a single term:

$$V = (D + E) \left(\frac{\text{FCF}}{D(1 - T_m)k_d + E(k_e) - (D + E)g} \right)$$

To complete the derivation of WACC in the denominator, divide the numerator and denominator by $(D + E)$. This will eliminate the $(D + E)$ expression on the left and place it in the denominator as a divisor. Distributing the term across the denominator, the result is the following equation:

$$V = \frac{\text{FCF}}{\frac{D}{D + E}(k_d)(1 - T_m) + \frac{E}{D + E}(k_e) - \frac{D + E}{D + E}(g)}$$

The expression in the denominator is the weighted average cost of capital (WACC) minus the growth in cash flow (g). Therefore, Equation B.1 can be rewritten as:

$$V = \frac{\text{FCF}}{\text{WACC} - g}$$

such that:

$$\text{WACC} = \frac{D}{D+E}(k_d)(1-T_m) + \frac{E}{D+E}(k_e)$$

Note how the after-tax cost of debt and the cost of equity are weighted by each security's *market-based* weight to enterprise value. This is why you should use market-based values, and not book values, to build the cost of capital. This is also why you should discount free cash flow at the weighted average cost of capital to determine enterprise value. Remember, however, that you can only use a constant WACC when leverage is expected to remain constant (i.e., debt grows as the business grows).²

ADJUSTED PRESENT VALUE

To determine enterprise value using adjusted present value, once again start with $V = D + E$ and multiply by a fraction equal to 1. This time, however, do not include the marginal tax rate in the fraction:

$$V = (D + E) \left(\frac{D(k_d) + \text{CF}_e - D(g)}{D(k_d) + \text{CF}_e - D(g)} \right)$$

Following the same process as before, convert each cash flow in the denominator to its present value times its expected return, and divide the fraction by $(D + E)/(D + E)$:

$$V = \frac{D(k_d) + \text{CF}_e - D(g)}{\frac{D}{D+E}(k_d) + \frac{E}{D+E}(k_e) - g}$$

Appendix C shows that if the company's interest tax shields have the same risk as the company's operating assets (as one would expect when the company maintains a constant capital structure), the fraction's denominator equals k_u , the unlevered cost of equity, minus the growth in cash flow (g). Make this substitution into the previous equation:

$$V = \frac{D(k_d) + \text{CF}_e - D(g)}{k_u - g}$$

² To see this restriction applied in a more general setting, see Miles and Ezzell, "Weighted Average Cost of Capital."

Next, focus on the numerator. Substitute the definitions of cash flow to debt and cash flow to equity, as we did earlier in this appendix:

$$V = \frac{\text{Interest} + \text{EBIT} - \text{Interest} - \text{Taxes} - \text{Net Investment} + D(g) - D(g)}{k_u - g}$$

In this equation, the two interest terms cancel and the two $D(g)$ terms cancel, so simplify by canceling these terms. Also insert $T_m(\text{Interest}) - T_m(\text{Interest})$ into the numerator of the expression:

$$V = \frac{\text{EBIT} - \text{Taxes} + T_m(\text{Interest}) - T_m(\text{Interest}) - \text{Net Investment}}{k_u - g}$$

Aggregate reported taxes and the negative expression for $T_m(\text{Interest})$ into all-equity taxes. Move the positive expression for $T_m(\text{Interest})$ into a separate fraction:

$$V = \frac{\text{EBIT} - [\text{Taxes} + T_m(\text{Interest})] - \text{Net Investment}}{k_u - g} + \frac{T_m(\text{Interest})}{k_u - g}$$

At this point, we once again have free cash flow in the numerator of the first fraction. The second fraction equals the present value of the interest tax shield. Thus, enterprise value equals free cash flow discounted by the unlevered cost of equity plus the present value of the interest tax shield:

$$V = \frac{\text{FCF}}{k_u - g} + \text{PV}(\text{Interest Tax Shield})$$

This expression is commonly referred to as adjusted present value.

In this simple proof, we assumed tax shields should be discounted at the unlevered cost of equity. This need not be the case. Some financial analysts discount expected interest tax shields at the cost of debt. If you do this, however, free cash flow discounted at the traditional WACC (defined earlier) and adjusted present value will lead to different valuations. In this case, WACC must be adjusted to reflect the alternative assumption concerning the risk of tax shields.

